Direct Control: (1) Targeted at high energy, **(2)** effective when installed, verified, and used properly, **(3)** Not vulnerable to human error

	Suspended load	Swinging Load	Fall from greater than 4 feet	Mobile equipment and workers on foot	Motor vehicle operations near (occupied vehicle)	Heavy rotating equipment	Surface temperatures (over 150 deg F)	Steam	Fire w/ substantial fuel source	Explosion	Excavation or trench 5 ft of depth or greater	Electrical contact w/ 50 volts or greater	Arc flash	High dose of toxic, chemical or radiation	Notes and Examples
Automatic collision avoidance or shutoff system				X	X										
Automatic fire suppression system									X	X					
Automatic shutoff system							X	X	X					X	
Blast blankets										X			X		
Bonding and grounding										X					Prevents static discharge of explosive chemicals.
Covers over holes			X												
De-energization w/ zero voltage check and grounding									X	X		X	X		**Example:** For potential wildfire ignition; gas operations; clearance including switching and tagging
Electrical insulation barriers												X	X		**Examples:** Pole guard, blankets, covers, line hoses
Energy isolation where LOTO is not required						X		X	X	X		X	X	X	**Examples:** Shutting a valve; **squeezing** and blocking *Does not include LOTO applied. Only applicable where LOTO is not required. Unplugging equipment,
Equipotential zones (EPZ)												X	X		Energy type: Induction
Excavation support system											X				**Examples:** Shoring, sheet piles, soldier piles
Fall arrest system*			X												**Examples:** Self retracting lifeline, netting system, ladder safety system. * Check integrity of the tie-off
Fall restraint system*			X												* Check integrity of the tie-off
Fuel source separation									X	X					**Example:** Static spray, wet rag, removal of combustibles
Gas detection monitoring															
Hard physical barrier - to stop equipment/vehicles				X						X					**Examples:** Jersey barriers and work vehicle positioning
Highway rated vehicle w/ seatbelt					X										**Examples:** Roll cage, seat belt and airbag systems.
Insulated or voltage–rated equipment and tools												X	X		**Examples:** Insulated sticks, booms, hot arms, hot line sticks.
Interlocks						X	X								**Examples:** Area curtains, 2 hand switches, pressure sensors on hydraulics. *Automatically detects and stops energy release.= or proximity.
Limitors/stabilizer that eliminates lateral motion		X													
Local exhaust ventilation														X	
Lock-out tag-out applied						X		X	X	X		X	X	X	Prevents unintended equipment operation or switch access. Isolate, test, and ground, Tags+ (tagging with other physical element to prevent energization)
Machine guards						X									
Mechanical pressure release devices										X					**Examples:** Rupture discs, shear pins.
Rated panel enclosure										X			X		For Explosion, due to combustible dust areas. Example: MCC, Metal clad switchgear. Panel enclosure is related to and installed for the hazard present.
Vehicle w/ Rollover Protection and Seatbelt					X										**Examples:** UTV, ATV, forklifts, bulldozers.
PPE Specialized - Arc rated and coverage per requirements													X		**Examples:** FR clothing, flash-rated equipment including suit, glasses, face shield, etc. Requirements based on system configuration and arc hazard analysis. Change made to include "requirement" rather than full coverage because SIFs are not happening when required coverage is present (to be verified)
PPE Specialized - Chemical or radiation rated														X	
PPE Specialized - Electrical (insulating gloves, sleeves, dielectric footwear,etc.)												X			
PPE Specialized - Heat resistant clothing							X								**Examples:** FR clothing, thermal suits, Kevlar sleeves, gloves, leathers for welding etc.
PPE Specialized- Steam suits w/ no skin exposed								X							
Radiation shielding														X	
Ratchet Straps or Tie-downs	X														Energy type: Gravity - Forklift or flatbed load lifts
Remote operation	X	X	X	X		X	X	X	X	X	X	X	X	X	**Examples:** Operations that take workers out of proximity. Cable cutting, racking, switching devices, drone operations, remote racking.
Sloping or benching											X				
Specialized Rated Containers														X	
Standard hard barrier exclusion zone - to stop people	X	X	X	X		X	X	X	X	X	X	X	X	X	**Examples:** Guardrails, handrails, concrete barriers.
Supplied air respirator														X	
Temperature Control Device							X								**Examples:** Active cooling system, awning over equipment.
Thermal surface insulation							X								
Tool forcing safe distance							X					X	X		**Example:** Hot sticks, load-break tools etc.
Tool tethers	X														Energy Type: Gravity - Falling object
Trench box											X				
Outriggers						X									